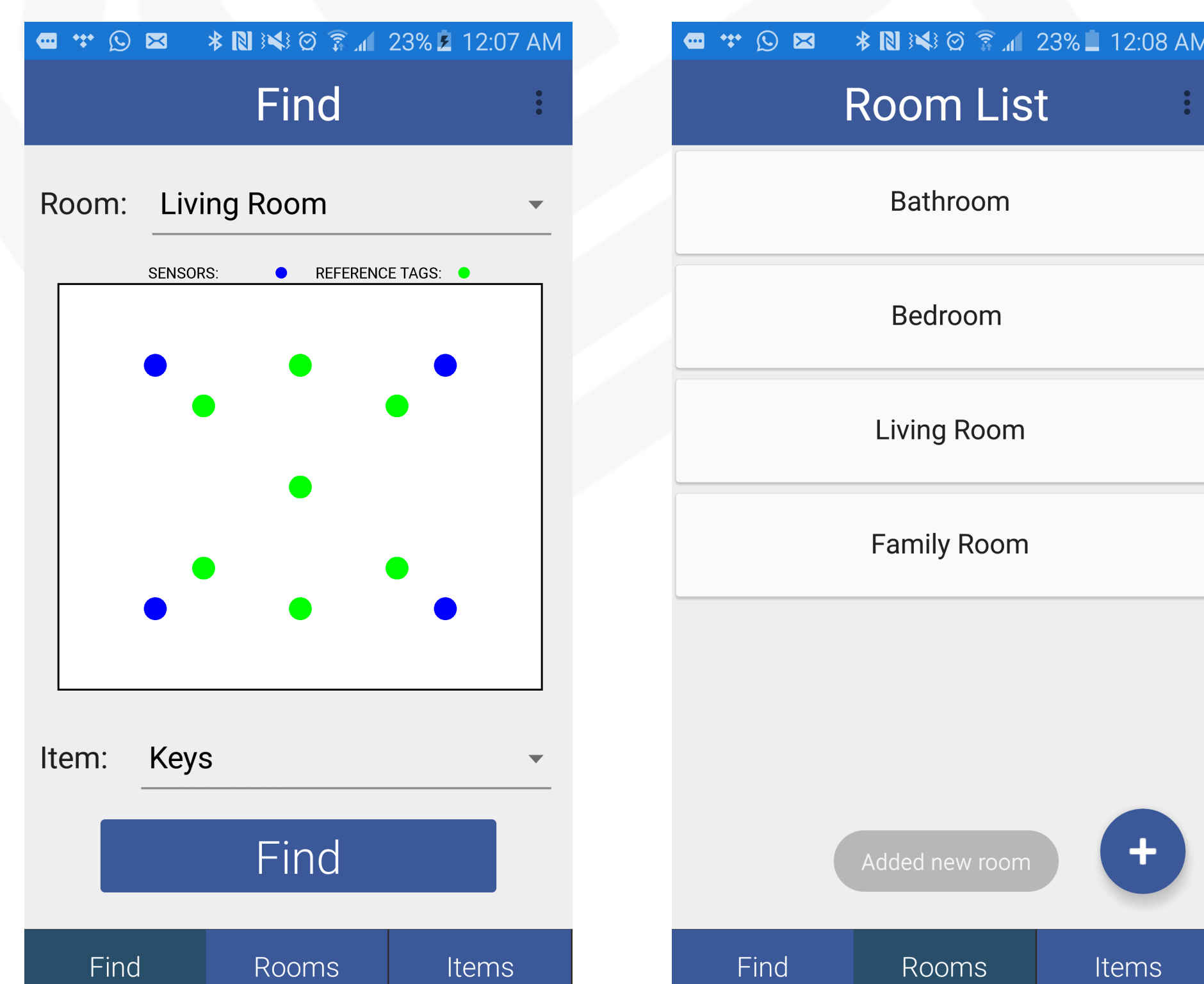


Our Motivation

- RFID technology is currently used for inventory, not tracking. (ex. using an access card to enter a library)
- Current trackers are bulky and expensive. We want a low cost, module free solution for missing items.
- To discover an alternative method to GPS tracking for locating items by incorporating the idea of the internet of things with emerging RFID technologies.

How it works



Android Application

- Allows a user to register RFID tags as items, set up rooms equipped with snapdragons and reference RFID tags, and then track their lost items on a map interface.
- Users can locate the room their item is currently in, or pinpoint exactly where an item specifically is inside a room.

Qualcomm Dragonboard 410c



- Used as server that listen for web server requests to activate the RFID readers and receive data back from them.
- Parses and formats the raw hexadecimal reader data and feed it back to the web server via HTTP.

Cottonwood RFID Readers

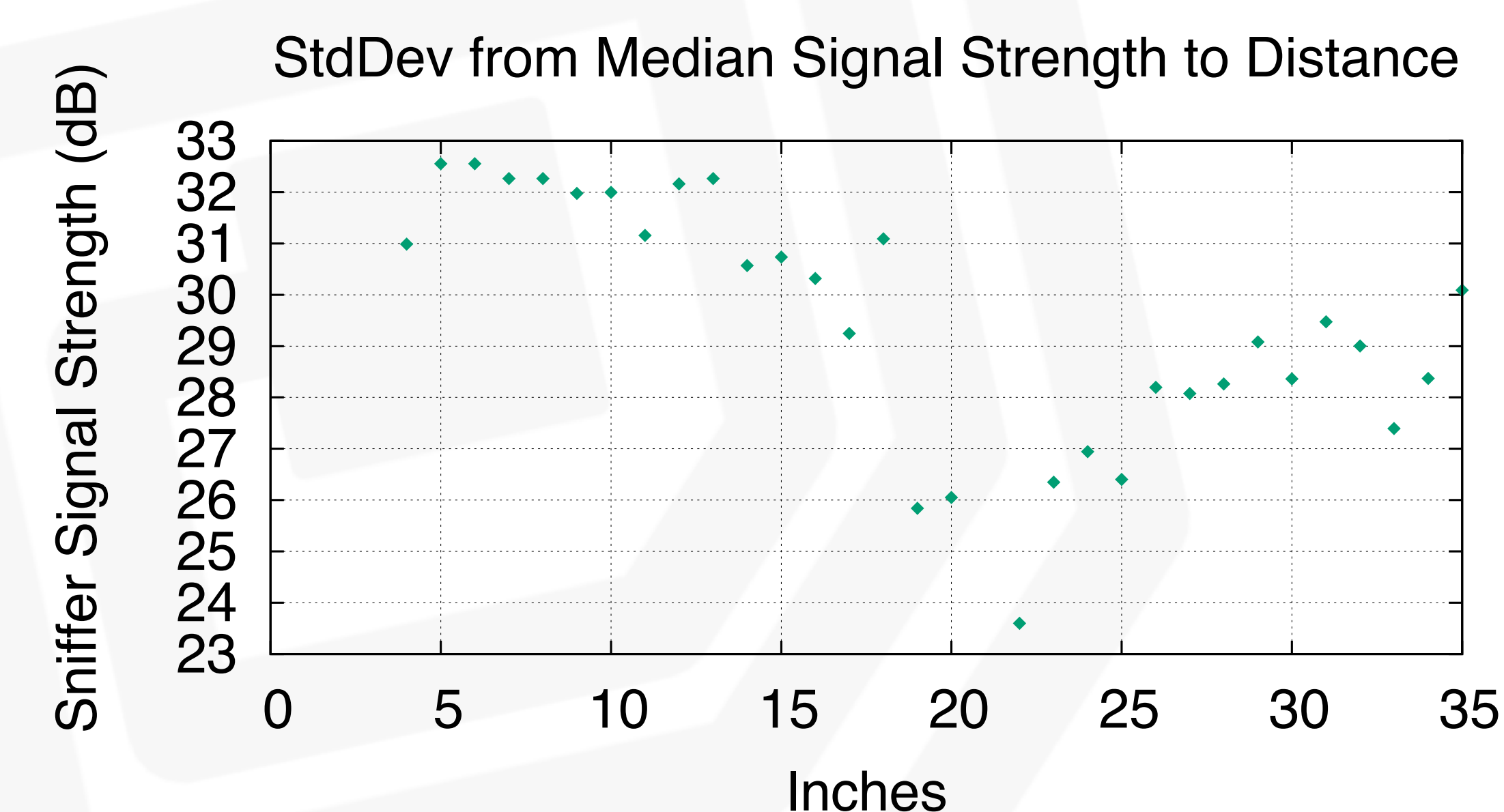


- Radiate a circularly polarized signal that picks up the presence of RFID stickers within its range.
- Associate an encoded signal strength value with each sticker, allowing for the possibility of relative positioning.

RFID Tags

- Passively store a unique identifier to be associated with a user's item.
- Powered by the electromagnetic energy via the reader, they transmit their stored information back to the reader in the form a radio wave.

RSSI-Distance Mapping



- The algorithm requires a mapping from RFID signal strength to approximate distance.
- The above graph shows the median absolute deviation of the signal strength returned by the RFID tag compared to the distance from the tag to the reader.
- From around 5 inches to around 20 inches, a very linear trend can be seen, correlating the two values. The zones used for the algorithms (x,y) plane were decided by this relationship.

Node JS

- Provides the REST API that enables the android application to send and receive data to our database
- Initiates interaction with the snapdragon processors.
- Runs the algorithm that aggregates data received from the Snapdragons and processes it accordingly.

Special Thanks:
Chandra Krintz
Tim Sherwood
Xiaolong Li
Luiza Timariu

